



How Are Continuous Motor Skills Learned and Represented?

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From Discrete Sequences to Continuous Skill Learning

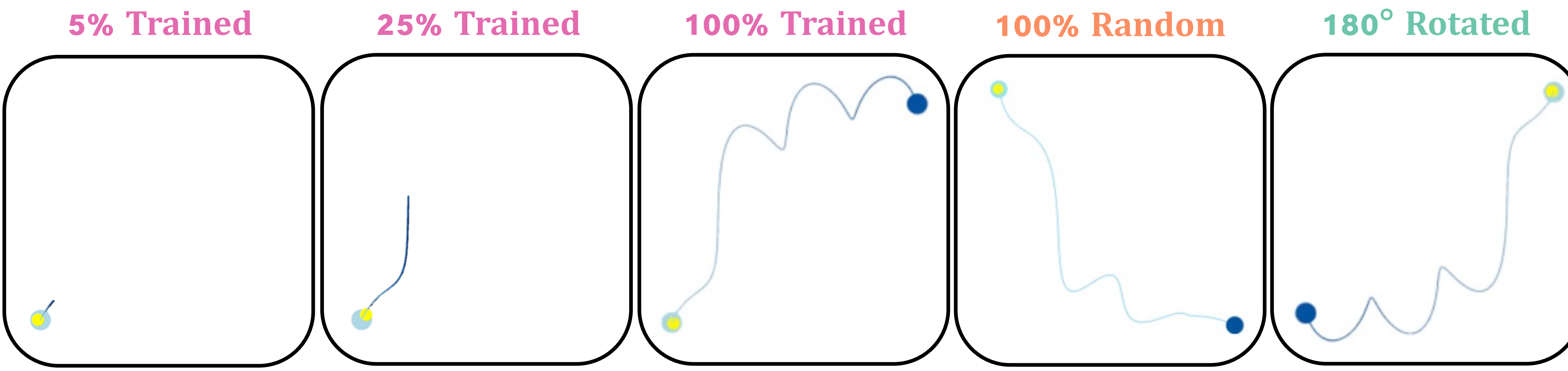
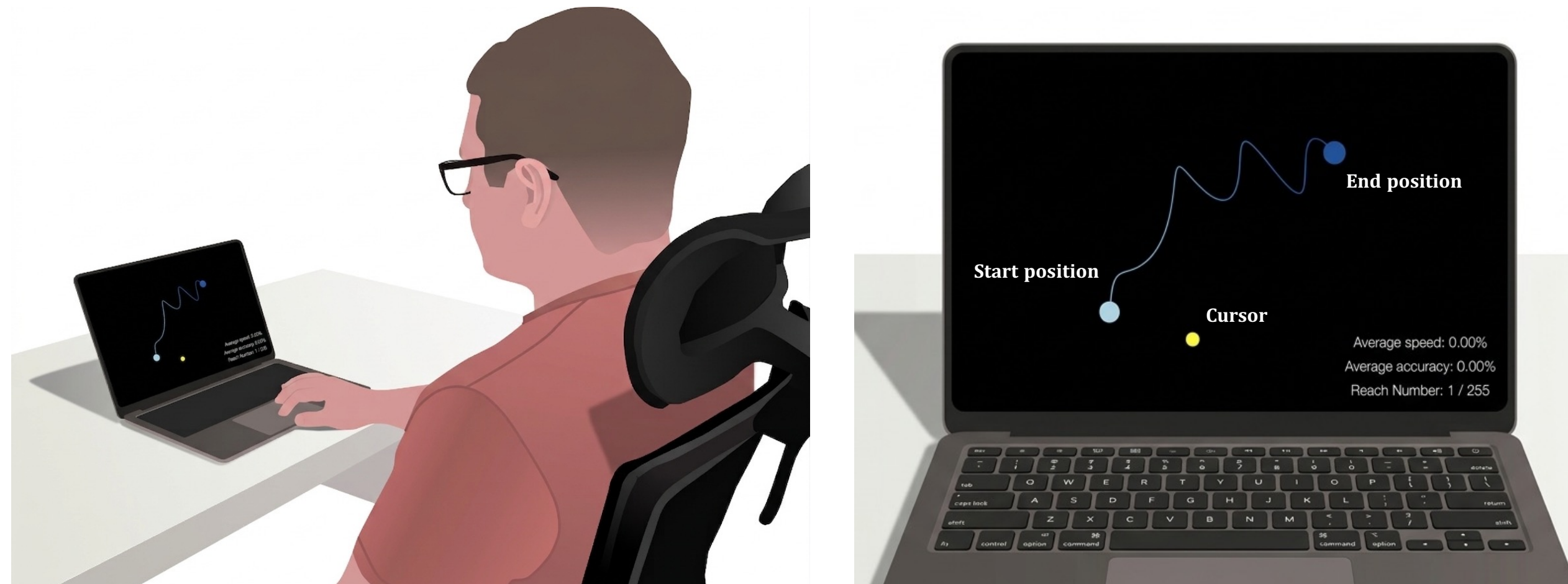
- Multiple processes contribute to the learning of **discrete, sequential skills**:
 - Sequence-specific gains that arise from improved predictions of upcoming elements¹
 - General skill improvements²
- Real-world motor skills require **continuous** actions with both speed and precision

We asked...

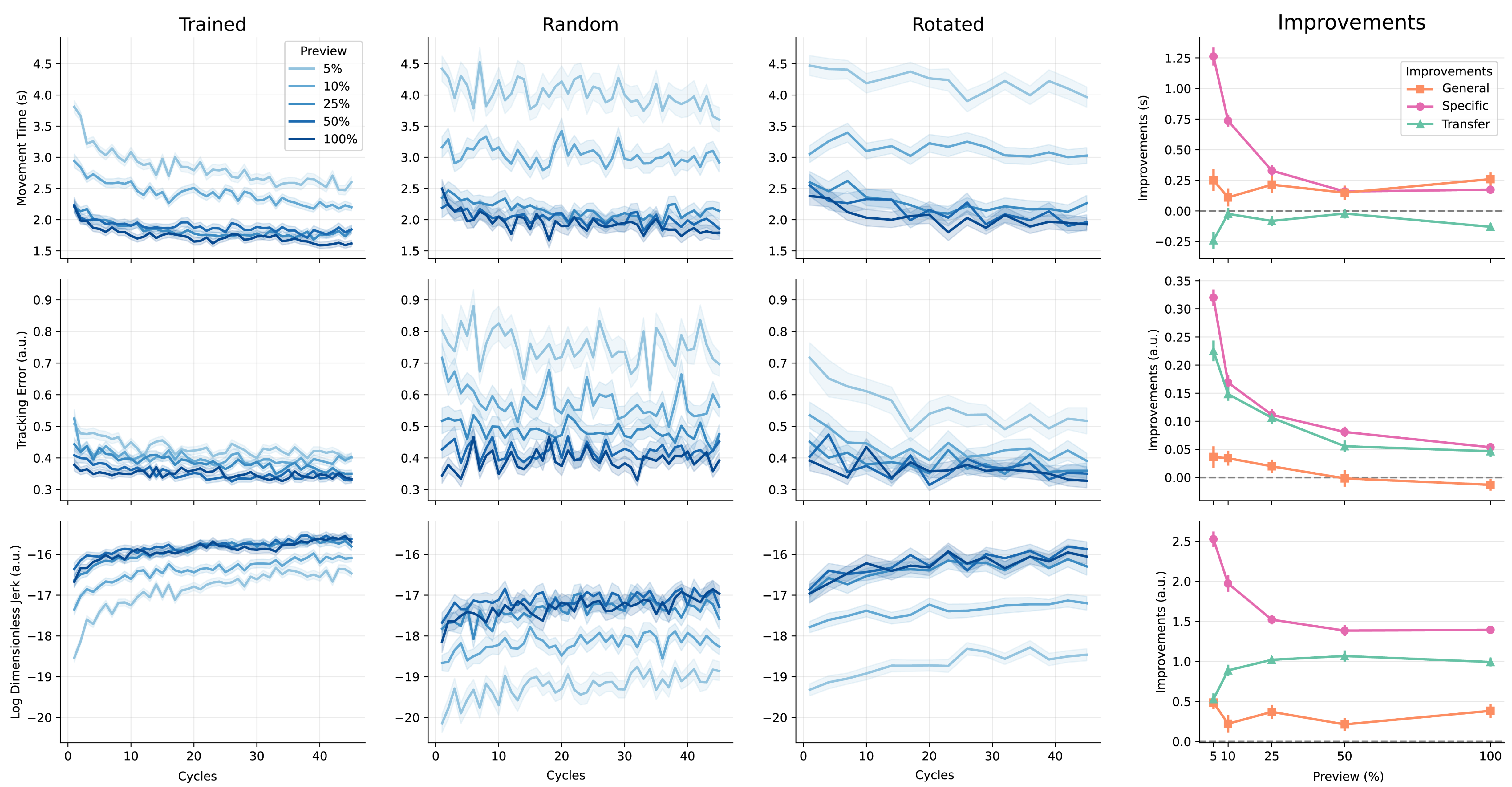
- How do continuous motor skills improve with practice?
- How are learned motor skills represented?

Evaluating continuous skill learning using a tracing task

- Task**: a web-based tracing task
- We varied how much of the upcoming trajectory is visible (**preview**) to control the extent of advance planning
- 3 trajectory types**: Trained, Random, Rotated



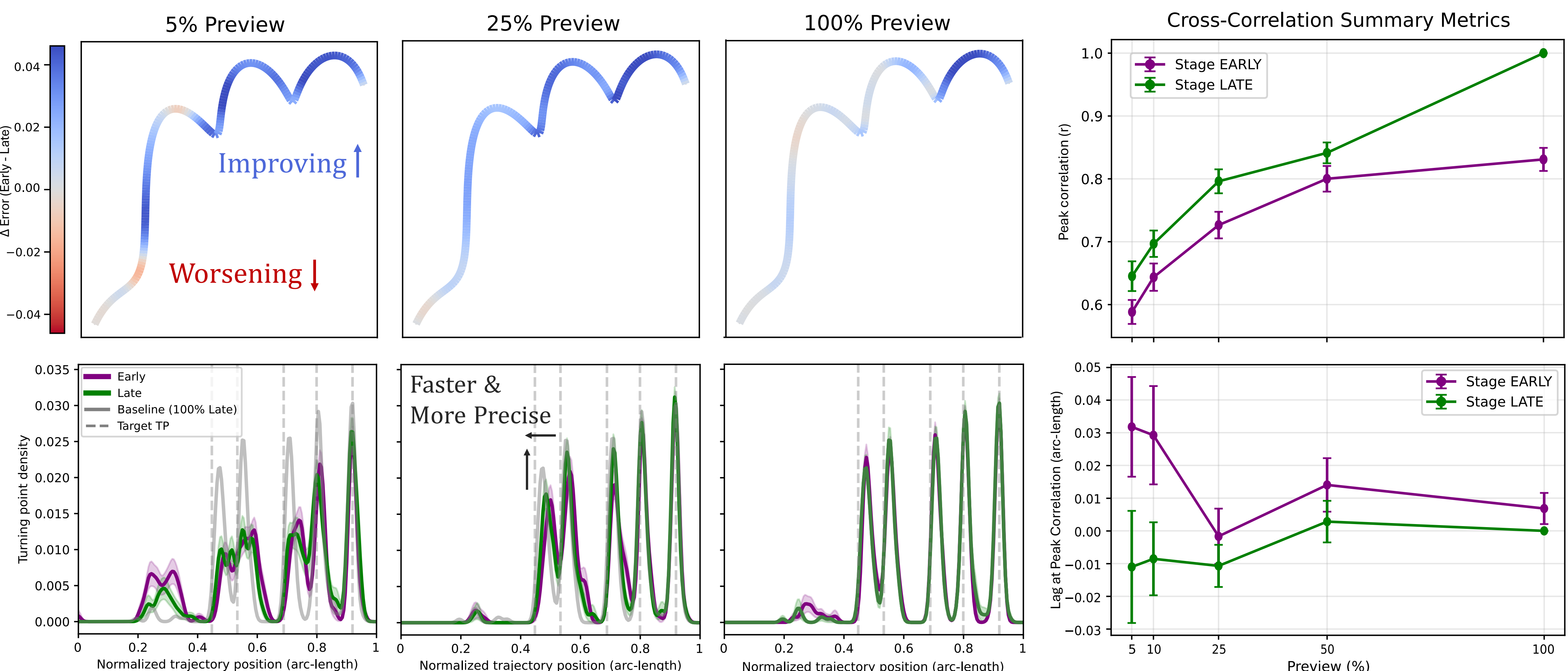
Continuous skill learning showed modest improvements in general planning/execution, but substantial trajectory-specific gains that leverage global shape representations



- ✓ **Trajectory-specific improvements:**
 - ✓ **Late Random - Late Trained**
 - Improved prediction of upcoming movements
- ✓ **General skill improvements:**
 - ✓ **Early Random - Late Random**
 - Improved motor efficiency
- ✓ **Shape representation:**
 - ✓ **Late Random - Late Rotated**
 - Learned continuous skills benefit from a global representation of the trajectory's shape

With practice, movements become faster, more accurate, and smoother (45 cycles / 240 Trials) ($N=300$; $n=60$ /group)

Trajectory-specific learning improved predictions of future movements, with strong modulation by preview conditions



More Precise / Feedforward movements with practice (TP Density: distribution of changes in movement direction)

Conclusions

- Continuous skill learning involves:
- Modest** general improvements in planning and execution
 - Substantial** trajectory-specific improvements driven by predictions of upcoming movements
 - Global shape** representations that facilitate transfer to similar but novel trajectories

Multiple learning processes identified in discrete skills extend to real-world continuous skills



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1. Bashford, L., Kobak, D., Diedrichsen, J., & Mehring, C. (2022). Motor skill learning decreases movement variability and increases planning horizon. *Journal of neurophysiology*, 127(4), 995-1006.
2. Kashefi, M., Diedrichsen, J., & Pruszynski, J. A. (2025). Motor sequence learning involves better prediction of the next action and optimization of movement trajectories. *Journal of Neuroscience*, 45(3).